

**Draft: new section to be inserted between Sec. 4 (Response functions)
and the numerical procedure**

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Note to co-authors. Equation numbers (5.x) below are provisional; references to Eqs. (2.2), (2.3), (4.6)–(4.12) are to the current version. Throughout, $E_\nu^{\text{hi}} \equiv 2 \text{ MeV}$ denotes the energy above which the flux is taken as a known input. Section 2 is unchanged: the equivalence with the dark-matter formulation, and Eq. (2.2), refer to the *whole* range of E_ν (as in DM to the whole range of v), where the boundary term of the integration by parts vanishes. The point of this new section is that the analysis is carried out on the *restricted* range $E_\nu < E_\nu^{\text{hi}}$, and there the separation must be made before integrating by parts.

5 Separation of the contributions below and above $E_\nu^{\text{hi}} = 2 \text{ MeV}$

5.1 Splitting the differential flux contribution

The differential rate of Eq. (4.6) is an integral over the whole neutrino energy range. Since the reactor spectrum above E_ν^{hi} is a measured input (Sec. 3), we separate the two contributions at the level of the differential flux $\phi(E_\nu)$, i.e. in Eq. (4.6) itself, before any integration by parts,

$$\frac{dR}{dE'}(E') = \int_0^{E_\nu^{\text{hi}}} dE_\nu \phi(E_\nu) \frac{\partial \mathcal{H}}{\partial E'}(E_\nu, E') + \int_{E_\nu^{\text{hi}}}^\infty dE_\nu \phi(E_\nu) \frac{\partial \mathcal{H}}{\partial E'}(E_\nu, E'). \quad (5.1)$$

It is convenient to introduce the response function acting on ϕ integrated over the i th data bin, in analogy with $R_{[E'_i, E'_{i+1}]}(E_\nu)$ of Eq. (4.12),

$$H_{[E'_i, E'_{i+1}]}(E_\nu) \equiv \int_{E'_i}^{E'_{i+1}} dE' \frac{\partial \mathcal{H}}{\partial E'}(E_\nu, E'), \quad \text{so that} \quad R_{[E'_i, E'_{i+1}]}(E_\nu) = \frac{dH_{[E'_i, E'_{i+1}]}(E_\nu)}{dE_\nu}, \quad (5.2)$$

which follows from Eq. (4.8). As noted below Eq. (4.7), $\partial \mathcal{H} / \partial E' \rightarrow 0$ as $E_\nu \rightarrow 0$ because the kinematic limit $E_R^{\text{max}}(E_\nu) \rightarrow 0$ collapses the integration domain; more precisely $H_{[E'_i, E'_{i+1}]}(E_\nu) = 0$ for $E_\nu \leq E_\nu^{\text{min}}$, with E_ν^{min} given by Eq. (4.2) evaluated at $E_R^{\text{min}} = E'_{\text{thr}}$. Hence

$$H_{[E'_i, E'_{i+1}]}(E_\nu) = \int_0^{E_\nu} dE R_{[E'_i, E'_{i+1}]}(E), \quad H_{[E'_i, E'_{i+1}]}(E_\nu^{\text{hi}}) = \int_{E_\nu^{\text{min}}}^{E_\nu^{\text{hi}}} dE_\nu R_{[E'_i, E'_{i+1}]}(E_\nu) = \sum_{j=1}^{N_{\text{int}}} R_{ij}, \quad (5.3)$$

the last equality following from the definition of R_{ij} , Eq. (5.3) of the current version. Integrating Eq. (5.1) over the i th bin as in Eq. (4.10) gives

$$\frac{N_i}{\mathcal{E}} = \underbrace{\int_0^{E_\nu^{\text{hi}}} dE_\nu \phi(E_\nu) H_{[E'_i, E'_{i+1}]}(E_\nu)}_{\equiv S_i} + \underbrace{\int_{E_\nu^{\text{hi}}}^\infty dE_\nu \phi(E_\nu) H_{[E'_i, E'_{i+1}]}(E_\nu)}_{\equiv h_i} + b_i. \quad (5.4)$$

Here h_i is the number of CE ν NS events per unit exposure in the i th bin produced by neutrinos with $E_\nu > E_\nu^{\text{hi}}$. This is the quantity plotted as the dotted green line in Fig. 2, and it coincides with the corresponding quantity of Ref. [6]. It is known because the flux above E_ν^{hi} is measured, and we assign to it a 3% uncertainty. Likewise S_i , the cyan dash-dotted line in Fig. 2, is the contribution of neutrinos with $E_\nu < E_\nu^{\text{hi}}$, which is the signal from which the low-energy spectrum is extracted.

5.2 Integration by parts on the restricted range

We now write S_i in terms of the integrated flux, repeating the step of Eq. (4.7) but on the finite interval $[0, E_\nu^{\text{hi}}]$. Using $\phi = -d\Phi/dE_\nu$ from Eq. (2.3) and Eq. (5.2),

$$\begin{aligned} S_i &= \left[-\Phi(E_\nu) H_{[E'_i, E'_{i+1}]}(E_\nu) \right]_0^{E_\nu^{\text{hi}}} + \int_0^{E_\nu^{\text{hi}}} dE_\nu \Phi(E_\nu) R_{[E'_i, E'_{i+1}]}(E_\nu) \\ &= -\Phi(E_\nu^{\text{hi}}) H_{[E'_i, E'_{i+1}]}(E_\nu^{\text{hi}}) + \int_0^{E_\nu^{\text{hi}}} dE_\nu \Phi(E_\nu) R_{[E'_i, E'_{i+1}]}(E_\nu). \end{aligned} \quad (5.5)$$

The lower limit contributes nothing since $H_{[E'_i, E'_{i+1}]}(0) = 0$, as in Eq. (4.7). The upper limit, however, does *not* vanish: in Eq. (4.7) the boundary term at $E_\nu \rightarrow \infty$ vanishes because $\Phi \rightarrow 0$, whereas here the domain ends at the finite energy E_ν^{hi} , where $\Phi(E_\nu^{\text{hi}}) \neq 0$ (Fig. 1) and $H_{[E'_i, E'_{i+1}]}(E_\nu^{\text{hi}}) \neq 0$. Using Eq. (5.3) to write $H_{[E'_i, E'_{i+1}]}(E_\nu^{\text{hi}}) = \int_0^{E_\nu^{\text{hi}}} dE_\nu R_{[E'_i, E'_{i+1}]}(E_\nu)$, the constant $\Phi(E_\nu^{\text{hi}})$ can be taken inside the integral, and

$$S_i = \int_0^{E_\nu^{\text{hi}}} dE_\nu \left[\Phi(E_\nu) - \Phi(E_\nu^{\text{hi}}) \right] R_{[E'_i, E'_{i+1}]}(E_\nu) = \int_0^{E_\nu^{\text{hi}}} dE_\nu \Delta\Phi(E_\nu) R_{[E'_i, E'_{i+1}]}(E_\nu) \quad (5.6)$$

where we have defined the integrated flux measured from the value at the splitting energy,

$$\Delta\Phi(E_\nu) \equiv \Phi(E_\nu) - \Phi(E_\nu^{\text{hi}}) = \int_{E_\nu}^{E_\nu^{\text{hi}}} d\tilde{E}_\nu \phi(\tilde{E}_\nu), \quad E_\nu \leq E_\nu^{\text{hi}}. \quad (5.7)$$

Thus, once the contribution of neutrinos above E_ν^{hi} is separated as in Eq. (5.4), the function of which the observable S_i is a convolution is not $\Phi(E_\nu)$ but $\Delta\Phi(E_\nu)$, the flux integrated between E_ν and E_ν^{hi} . This is the quantity that the numerical procedure of the next section determines. Note that the two functions differ by the constant $\Phi(E_\nu^{\text{hi}})$, which is fixed by the same measured high-energy flux that fixes h_i ; for the spectrum of Fig. 1, $\Phi(2 \text{ MeV}) = 4.6 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$, i.e. 24% of $\Phi(E_\nu^{\text{min}})$, so the distinction is numerically important.

5.3 Properties of $\Delta\Phi$ and validity of the results of Sec. 2

All the properties of Φ used in Sec. 2 are inherited by $\Delta\Phi$:

- $\Delta\Phi(E_\nu) \geq 0$ on $[E_\nu^{\text{min}}, E_\nu^{\text{hi}}]$, and $\Delta\Phi$ is non-increasing, because Φ is non-increasing and $\Phi(E_\nu^{\text{hi}})$ is a constant. The admissible set is therefore the same convex cone as for Φ , defined by Eq. (5.8) of the current version.
- Consequently the results quoted in Sec. 2 apply verbatim: by the Fenchel–Eggleston theorem applied to the convex hull generated by the response functions, the best fit is a piecewise constant function of E_ν with at most $d - 1$ downward steps, and the pointwise confidence band is constructed exactly as described there and in Appendix C. Since $\Delta\Phi$ and Φ differ by a constant, the two functions have identical step structure.
- Working with $\Delta\Phi$ has the additional advantage that the condition $\Delta\Phi \geq 0$ is precisely the statement that $\Phi(E_\nu) \geq \Phi(E_\nu^{\text{hi}})$ for $E_\nu < E_\nu^{\text{hi}}$, i.e. it enforces the monotonicity of Φ across the splitting energy, which a fit of Φ subject only to $\Phi_j \geq 0$ would not impose.
- The integrated flux itself is recovered by the rigid shift $\Phi(E_\nu) = \Delta\Phi(E_\nu) + \Phi(E_\nu^{\text{hi}})$, so that any confidence band obtained for $\Delta\Phi$ translates into a band for Φ displaced by the known constant $\Phi(E_\nu^{\text{hi}})$ (up to its 3% uncertainty, treated as for h_i).

5.4 Ansatz and predicted number of events

We divide $[E_\nu^{\min}, E_\nu^{\text{hi}}]$ into N_{int} equal intervals in which $\Delta\Phi$ takes a constant value $\Delta\Phi_j$ (this replaces Eq. (5.1) of the current version and Fig. 4),

$$\Delta\Phi(E_\nu) = \sum_{j=1}^{N_{\text{int}}} \Delta\Phi_j \left[\Theta(E_\nu - E_\nu^j) - \Theta(E_\nu - E_\nu^{j+1}) \right]. \quad (5.8)$$

With Eq. (5.6) and the response matrix R_{ij} of Eq. (5.3) of the current version — which is *unchanged* — the signal becomes $S_i = \sum_j R_{ij} \Delta\Phi_j$, and the predicted number of events in the i th bin, Eq. (5.4) of the current version, reads

$$N_i = \mathcal{E} \left[\sum_{j=1}^{N_{\text{int}}} R_{ij} \Delta\Phi_j + h_i + b_i \right]. \quad (5.9)$$

The parameters $\{\Delta\Phi_j\}$ are determined by minimizing the Neyman χ^2 ,

$$\chi^2 = \sum_i \frac{1}{\mathcal{E} O_i} \left[\mathcal{E} \left(\sum_{j=1}^{N_{\text{int}}} R_{ij} \Delta\Phi_j + h_i + b_i \right) - \mathcal{E} O_i \right]^2, \quad (5.10)$$

subject to

$$\Delta\Phi_j \geq 0, \quad \Delta\Phi_{j+1} \leq \Delta\Phi_j, \quad j = 1, \dots, N_{\text{int}} - 1. \quad (5.11)$$

Everything that follows — the choice of N_{int} , the profiling over $\{\Delta\Phi_k\}_{k \neq j}$, the degeneracy band, and the Monte Carlo calibration of $\Delta\chi_{\text{min}}^2$ described in Appendix C — is unchanged, with Φ_j replaced throughout by $\Delta\Phi_j$.

Changes implied elsewhere in the paper

1. Sec. 2 unchanged: it establishes the DM/neutrino equivalence over the whole E_ν range, where the boundary term vanishes. A forward reference can be added, e.g. “the restriction to $E_\nu < 2 \text{ MeV}$ is treated in Sec. 5”.
2. Sec. 4 unchanged (Eqs. (4.1)–(4.12)). Eq. (4.9) remains the whole-range statement.
3. Current Eqs. (5.1), (5.2), (5.4), (5.7), (5.8) are replaced by Eqs. (5.8)–(5.11) above; Fig. 4 relabelled $\Phi_j \rightarrow \Delta\Phi_j$.
4. The sentence defining h_i below Fig. 2 is now consistent with Eq. (5.9) and needs no change; Fig. 2 is unchanged.
5. Flux figures (Figs. 7–11): either relabel the vertical axis as $\Delta\Phi(E_\nu)$ and shift the theoretical curves down by $\Phi(2 \text{ MeV})$, or add $\Phi(2 \text{ MeV})$ to the bands and keep the axis as $\Phi(E_\nu)$, as noted in Sec. 5.3. To be decided.
6. The comparison with the integrated band of Ref. [6] should be checked against whichever of the two conventions is adopted in item 5.
7. No fit and no confidence band has to be recomputed.